

Alex Lewandowski

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I am working to build an agent that effectively uses its experience to continually adapt its behaviour.

Education

Ph.D. in Computing Science

UNIVERSITY OF ALBERTA

2019 – Present

- » Supervisors: Dale Schuurmans, Marlos C. Machado
- » Research topics: reinforcement learning, continual learning, meta-learning
- » Anticipated graduation: 04/2026

M.Sc. in Statistics

UNIVERSITY OF ALBERTA

2016 – 2018

- » Supervisors: Ivor Cribben, Rohana Karunamuni
- » Thesis: Recurrent and Bayesian Kernel Learning for Small Data with Applications to Neuroimaging

Honours Bachelor in Mathematics

UNIVERSITY OF WATERLOO

2012 – 2016

- » Major: Mathematical Economics

Papers & Preprints

The World is Bigger: Interaction and Learning Within a World

06/2025

- » **A. Lewandowski**, A. Ramesh, E. Meyer, S. Kumar, D. Schuurmans, M. C. Machado
- » Reinforcement Learning and Decision Making (RLDM)

Plastic Learning with Deep Fourier Features

04/2025

- » **A. Lewandowski**, D. Schuurmans, M. C. Machado
- » International Conference on Learning Representations (ICLR)

Learning Continually by Spectral Regularization

04/2025

- » **A. Lewandowski**, M. Bortkiewicz, S. Kumar, A. György, D. Schuurmans, M. Ostaszewski, M. C. Machado
- » International Conference on Learning Representations (ICLR)

The Need for a Big World Simulator: A Scientific Challenge for Continual Learning

08/2024

- » S. Kumar, H. J. Jeon, **A. Lewandowski**, B. Van Roy
- » Finding The Frame Workshop at RLC (Oral Presentation)

Directions of Curvature as an Explanation for Loss of Plasticity

11/2023

- » **A. Lewandowski**, H. Tanaka, D. Schuurmans, M. C. Machado
- » Preprint

Reinforcement Teaching

06/2023

- » C. Muslimani*, **A. Lewandowski***, D. Schuurmans, M. Taylor, J. Luo
- » Transactions on Machine Learning Research (TMLR)

ZORB: A Derivative-Free Backpropagation Algorithm for Neural Networks

12/2020

- » V. Ranganathan, **A. Lewandowski**
- » Beyond Backpropagation Workshop at NeurIPS (Oral Presentation)

Work Experience

Research Associate (Part-time), Noah's Ark Lab

HUAWEI TECHNOLOGIES CANADA CO., LTD.

2022 – 2023

- » Supervisor: Jun Luo
- » Developed a reinforcement learning framework for meta-learning to improve training speed of new agents

Research Associate (Full-time), Noah's Ark Lab

HUAWEI TECHNOLOGIES CANADA CO., LTD.

2021 – 2022

» Supervisor: Jun Luo

» Designed an evaluation methodology for generalization of reinforcement learning agents, applied to autonomous vehicles

Teaching Assistant, Department of Computer Science

UNIVERSITY OF ALBERTA

2019 – Present

» Facilitated discussion groups and developed worksheets for a new reinforcement learning course

» Provided mentorship and guidance for student projects in foundations of information retrieval

Teaching Assistant, Department of Mathematical and Statistical Sciences

UNIVERSITY OF ALBERTA

2016 – 2018

» Conducted help sessions in Applied Statistics, Statistics I/II, Applied Regression Analysis and Time Series Analysis.

» Offered personalized academic support to students in all first and second-year classes in mathematics and statistics at the Decima Robinson Support Centre

Honors & Awards

Alberta Innovates Graduate Student Scholarship

University of Alberta, Department of Computing Science

2022–2024

President's Doctoral Prize of Distinction

University of Alberta, Department of Computing Science

2021–2024

NSERC Postgraduate Scholarships – Doctoral

University of Alberta, Department of Computing Science

2021–2024

Josephine Mitchell Scholarship

University of Alberta, Department of Mathematical and Statistical Sciences

2017–2018

Profiling Alberta's Graduate Students Award

University of Alberta, Department of Mathematical and Statistical Sciences

2017

Queen Elizabeth II Graduate Scholarship

University of Alberta, Department of Mathematical and Statistical Sciences

2016

President's Scholarship

University of Waterloo

2012

Skills

Languages

Python (Jax, PyTorch), Julia, VC (git)

Service

Organizer

RLC 2025 (Local Chair), Finding the Frame Workshop at RLC 2025, Openmind Research Institute Banff Retreat 2023-2024, ICLR SSBM Social 2021, NeurIPS SSBM Social 2020, ICLR Reinforcement Learning Social 2020

Reviewer

NeurIPS 2021-2025 (Top Reviewer: 2022, 2025), ICLR 2020-2025 (Outstanding Reviewer: 2021, 2022), ICML 2021-2024 (Outstanding Reviewer: 2022), RLC 2024-2025, CoLLAs 2025, Finding the Frame Workshop at RLC 2024, ICML RL4RealLife Workshop 2021, NeurIPS Offline RL Workshop 2021, OPTRL Workshop at NeurIPS 2019,

University

Academic Director at Computing Science Graduate Student Association 2022–2023